```
function main()
nl=5;
N=20;
R=1.5;
sz=1;
I=300;
mo=4*pi*1e-7;
km=mo*I/(4*pi);
rw=0.2;
plot_option=true;
ds=0.1;
[Px,Py,Pz,dx,dy,dz] = espiras(sz,nl,N,R);
campoB(ds,km,Px,Py,Pz,dx,dy,nl,N,rw,plot_option);
end
%% FUNCIÓN ESPIRAS
function [Px,Py,Pz,dx,dy,dz] = espiras(sz,nl,N,R)
dtheta = 2*pi/N;
ang = 0:dtheta:(2*pi-dtheta); % Vector de ángulos
s=1;
for i=1:nl
    Px(s:s+N-1) = R*cos(ang);
    Py(s:s+N-1) = R*sin(ang);
    Pz(s:s+N-1) = -(nl-1)/2*sz + (i-1)*sz; % Centrar la espira
    dx(s:s+N-1) = -Py(s:s+N-1)*dtheta;
    dy(s:s+N-1) = Px(s:s+N-1)*dtheta; % Diferencial de línea
    s = s+N;
end
dz = zeros(1,N*nl);
figure(1)
quiver3(Px,Py,Pz,dx,dy,dz,0.5,'-r','LineWidth',2)
view(-34,33)
end
%% FUNCIÓN CAMPO B
function campoB(ds,km,Px,Py,Pz,dx,dy,nl,N,rw,plot_option)
z = -5.2:ds:5.2;
if plot_option
    x = z;
    y = z;
else
    x = -0.1:0.01:0.1; % Grid de puntos campo B
    y = -0.1:0.01:0.1;
end
1
Lx=length(x); Ly=length(y); Lz=length(z);
dBx=zeros(Lx,Ly,Lz,'single'); % single para ahorrar memoria
dBy=zeros(Lx,Ly,Lz,'single');
dBz=zeros(Lx,Ly,Lz,'single');
for i=1:Lx
    for j=1:Ly
        for k=1:Lz
```

```
            for l=1:nl*N
                rx = x(i)-Px(l); % Punto campo menos punto físico
                ry = y(j)-Py(l);
                rz = z(k)-Pz(l);
                r = sqrt(rx^2+ry^2+rz^2+rw^2);
                r3 = r^3;
                dBx(i,j,k) = dBx(i,j,k) + km*dy(l)*rz / r3;
                dBy(i,j,k) = dBy(i,j,k) - km*dx(l)*rz / r3;
                dBz(i,j,k) = dBz(i,j,k) + km*(dx(l)*ry - dy(l)*rx) / r3;
            end
        end
    end
end
if plot_option
    Bmag = sqrt(dBx.^2+dBy.^2+dBz.^2);
    centery = round(Ly/2);
    Bx_xz = squeeze(dBx(:,centery,:));
    Bz_xz = squeeze(dBz(:,centery,:));
    Bxz = squeeze(Bmag(:,centery,:));
    figure(2); hold on
    pcolor(x,z,(Bxz').^(1/3)); shading interp; colormap jet; colorbar
    h1 = streamslice(x,z,Bx_xz',Bz_xz',3);
    set(h1,'Color',[0.8 1 0.9]);
    xlabel('x'); ylabel('z');
    title('Campo magnético generado por un solenoide');
end
end
```